

Local AI Deployment in Healthcare Enterprise Operations: Preserving Sovereignty, Data Privacy, and Continuity in Prior Authorization

Introduction and Topic Concept

The integration of Artificial Intelligence (AI) into healthcare operations represents a fundamental paradigm shift in clinical decision-making and health system administration. Specifically, within Treatment, Payment, and Operations (TPO)—most notably prior authorization workflows—AI offers the potential to not only alleviate immense administrative burdens but also significantly decrease delay times for patients awaiting critical medical care (Mazzolini). However, while existing industry discourse places heavy emphasis on algorithmic capabilities, insufficient attention has been directed toward how AI models are deployed, controlled, and governed. This paper evaluates the operational trade-offs between cloud-hosted Model-as-a-Service (MaaS) APIs and air-gapped, local AI deployments for TPO functions, showcasing a sovereign architecture engineered specifically for prior authorization.

Rather than relying on public cloud APIs from external providers, we propose a strategic transition toward local deployment of Large Language Models (LLMs). Local AI deployment involves hosting open-weight models entirely within an organization's secure, air-gapped infrastructure. By shifting the deployment paradigm from remote cloud calls to local inference engines, healthcare entities—and specialized health technology platforms like Qwen—can achieve complete data sovereignty, eliminate third-party data egress, and insulate core administrative workflows from external service volatility (Hou et al.).

Structural Bottlenecks and Governance Risks in Prior Authorization

Prior authorization remains one of the most persistent operational choke points across the United States healthcare sector. Healthcare providers receive an average of 50 authorization requests per week, consuming over 12 staff hours in manual documentation and review—resources that could otherwise be directed toward direct patient care (Mazzolini). The consequences of these administrative delays extend far beyond administrative friction; patients bear the tangible burden of stalled care, experiencing heightened anxiety, severe distress, treatment abandonment, permanent impairment, or even preventable mortality (Mazzolini).

While recent literature explores how AI algorithms can automate clinical documentation and decision support (Nguyen), the regulatory and architectural mechanisms required to govern sensitive health data remain underexplored. Processing Protected Health Information (PHI) under Health Insurance Portability and Accountability Act (HIPAA) rules introduces severe legal challenges regarding data privacy and institutional compliance when using commercial cloud endpoints (Rezaeikhonakdar).

For health-tech enterprises handling health operations, routing enterprise workflows through external cloud LLM APIs introduces three critical vulnerabilities.

First, commercial cloud endpoints expose proprietary business logic and sensitive clinical records to data leakage. Empirical research confirms that LLMs memorize training and fine-tuning data verbatim rather than merely learning abstract concepts (Tishler). Consequently, adversarial prompting or secondary extraction techniques can force cloud-hosted models to output confidential patient records, clinical guidelines, and proprietary business rules (Tishler).

Second, building core operational infrastructure around closed commercial APIs creates high-friction vendor dependencies. Unannounced modifications to API pricing structures, usage throttles, or abrupt base-model deprecation ("rug-pulling") can instantly destabilize enterprise workflows, forcing costly systemic re-alignment (Hou et al.).

Third, fine-tuning proprietary cloud models leaves the resulting weights locked within the vendor's cloud ecosystem. Organizations are unable to export, audit, or permanently retain their fine-tuned model weights, forfeiting long-term intellectual property control (Hou et al.).

The Local Deployment Imperative and Infrastructure Viability

Deploying open-weight LLMs locally on-premise or within a private, air-gapped cloud directly resolves these vulnerabilities. By maintaining physical ownership of model weights, organizations guarantee total data isolation, satisfying stringent HIPAA compliance parameters (Rezaeikhonakdar). Furthermore, local deployment converts unpredictable per-token operational expenditures into stable capital infrastructure investments.

Crucially, hosting models locally no longer requires compromising on intelligence or capability. Given the appropriate infrastructure, modern open-weight systems rival closed-source frontier clouds; for example, open models like Kimi K3 demonstrate reasoning capabilities that approach top-tier commercial endpoints (Kimi Team), demonstrating that local deployment is both scalable and capable for full enterprise production and production-grade clinical reasoning.

Strategic Proposal for Cotiviti

To further strengthen Cotiviti's end-to-end prior authorization solution and FHIR Gateway architecture, Cotiviti should adopt an air-gapped, locally deployed LLM framework—leveraging open-weight models—as a core design principle for its generative AI operations. This local deployment strategy delivers three transformative advantages for Cotiviti.

First, operating open-weight LLMs entirely within Cotiviti-controlled infrastructure achieves zero data egress and uncompromising HIPAA compliance. This reinforces Cotiviti's commitment to responsible AI while guaranteeing to healthcare clients that patient data is never exposed to third-party model training pipelines or external API vulnerabilities (Rezaeikhonakdar).

Second, local model deployment safeguards Cotiviti's proprietary payment integrity logic. Cotiviti's clinical rules, medical policies, and data mining algorithms represent high-value trade secrets; utilizing local fine-tuning ensures that custom prompt engineering and policy datasets

are never absorbed by commercial cloud vendors or extracted via verbatim recall vulnerabilities (Tishler).

Finally, local deployment secures operational sovereignty, performance, and cost predictability by decoupling Cotiviti's prior authorization engines from commercial cloud API pricing structures and deprecation schedules. Local inference provides scalable throughput and complete control over model checkpoints, ensuring that Cotiviti's automated workflows remain resilient, cost-effective, and fully under human specialist governance.

Works Cited

Hou, Xinyi, et al. "LLM Applications: Current Paradigms and the Next Frontier." *arXiv*, 9 Oct. 2025, <https://doi.org/10.48550/arXiv.2503.04596>.

Kimi Team. "Kimi K3: Open Frontier Intelligence." *arXiv*, 27 July 2026, <https://doi.org/10.48550/arXiv.2607.24653>.

Mazzolini, Chris. "Prior Authorization: How It Evolved, Why It Burdens Physicians and Patients, and the Promise of AI." *Medical Economics*, 24 Apr. 2025, www.medicaleconomics.com/view/prior-authorization-history-burden-ai-future.

Nguyen, Tan Tho. "Transforming Prior Authorization through Artificial Intelligence: A National Framework for Reducing Administrative Burden and Improving Patient Access." *International Journal of AI, BigData, Computational and Management Studies*, vol. 7, no. 3, 2026, pp. 17–28, <https://doi.org/10.63282/3050-9416.IJAIBDCMS-V7I3P103>.

Rezaeikhonakdar, Delaram. "AI Chatbots and Challenges of HIPAA Compliance for AI Developers and Vendors." *The Journal of Law, Medicine & Ethics*, vol. 51, no. 4, 2023, pp. 988–95, <https://doi.org/10.1017/jme.2024.15>.

Tishler, Sarah. "Verbatim Recall in LLMs: A New Study Raises Important Questions for Trade Secret Protection." *Beck Reed Riden LLP*, 13 Apr. 2026, beckreedriden.com/verbatim-recall-in-llms-a-new-study-raises-important-questions-for-trade-secret-protection/.